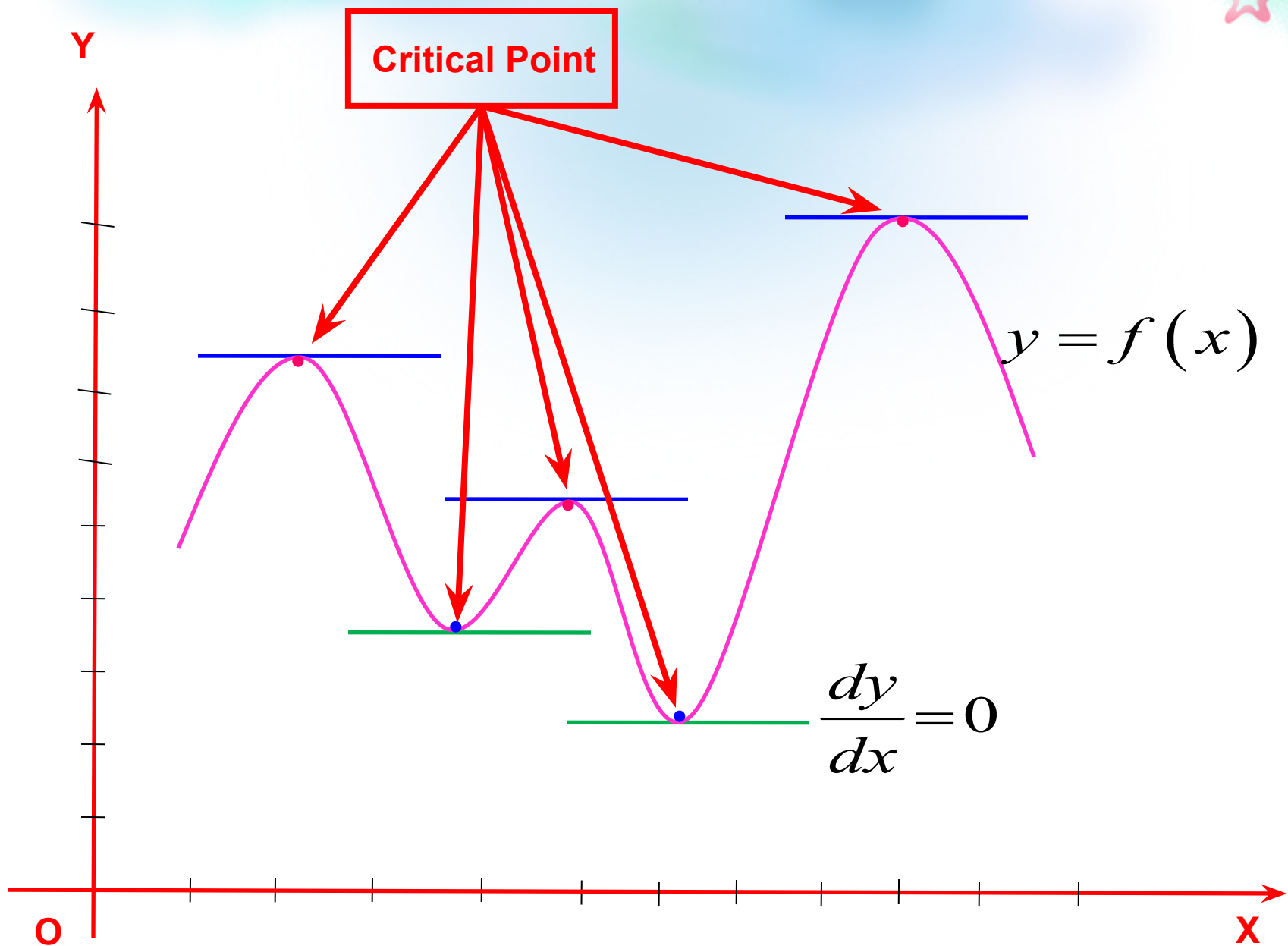
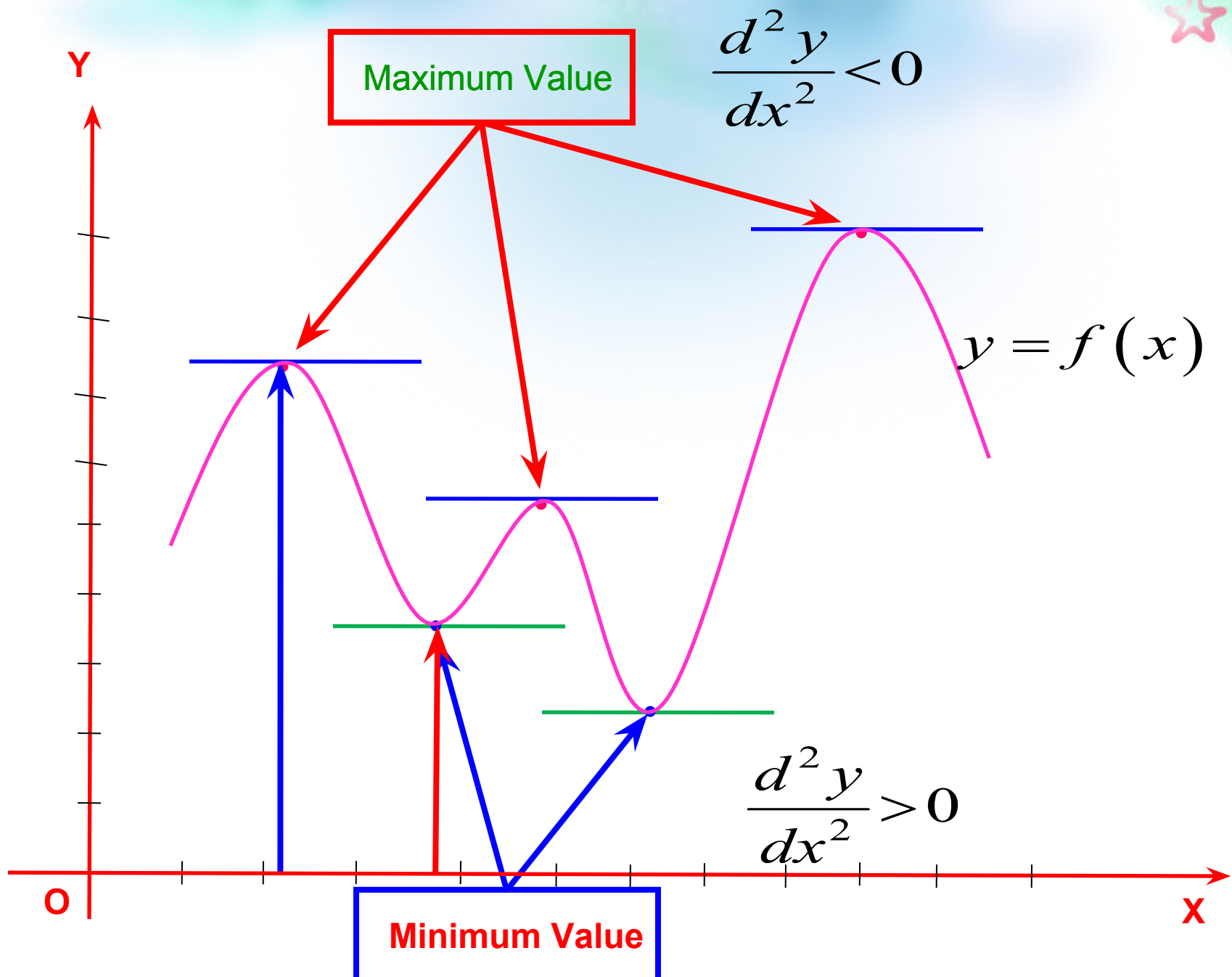
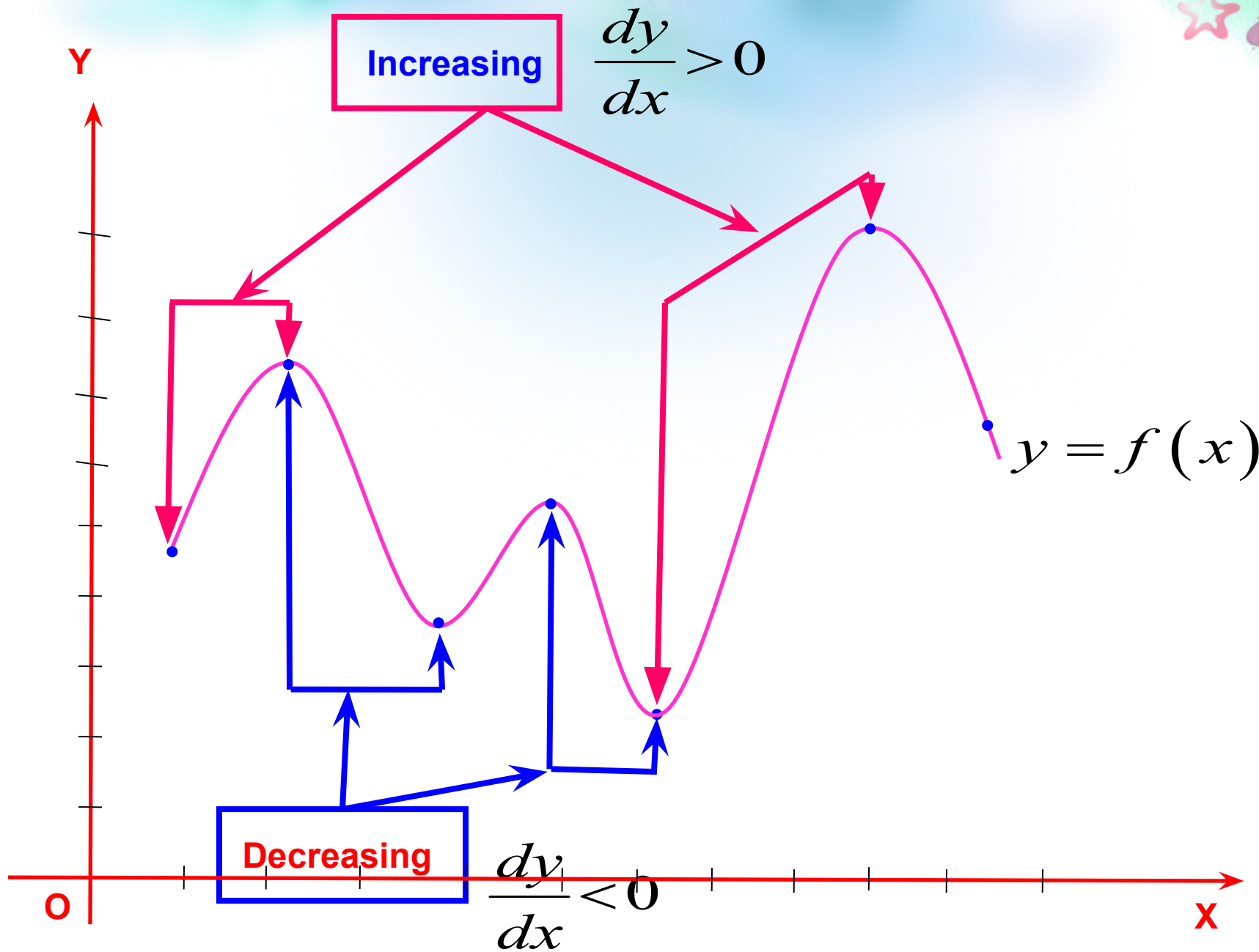


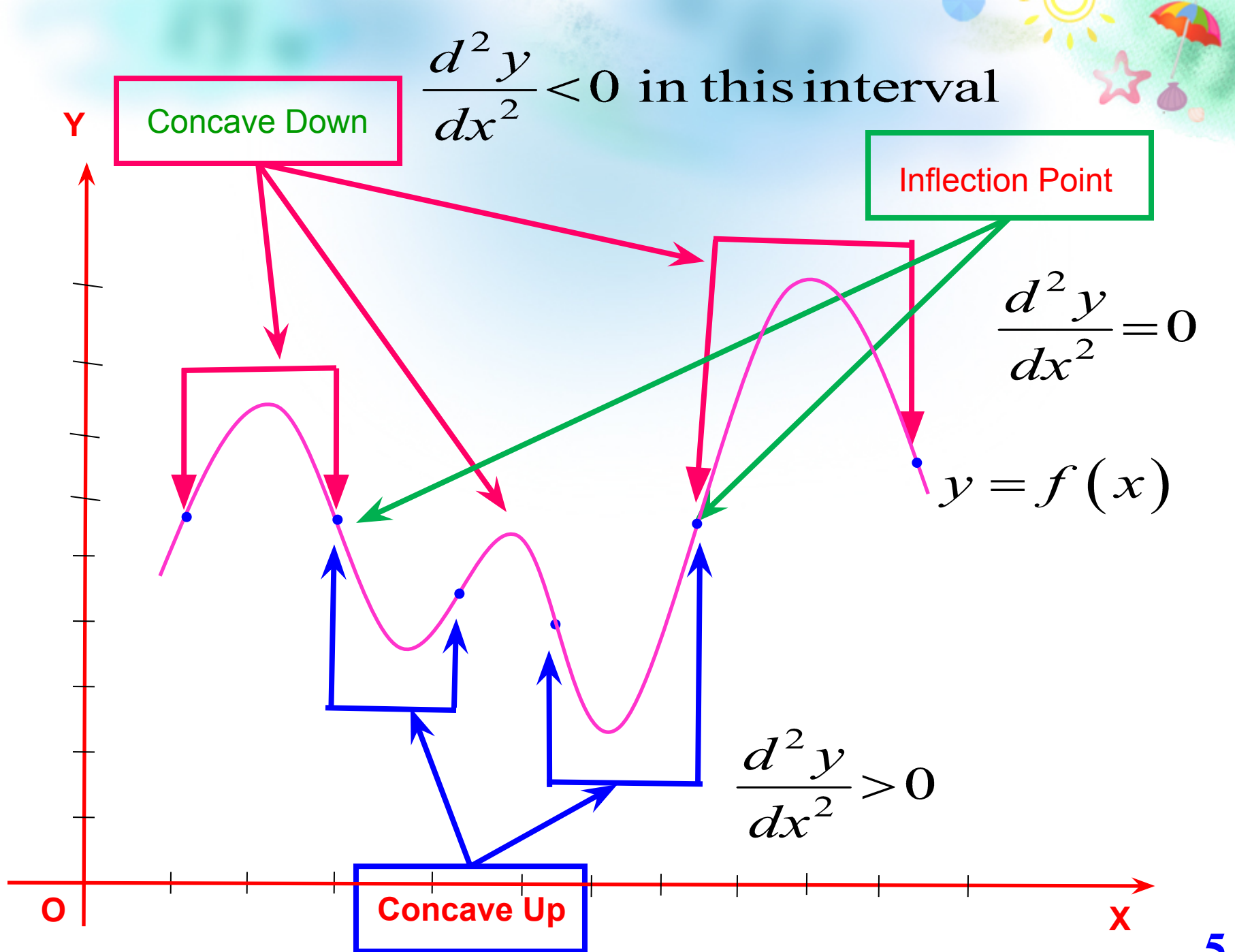


Welcome To Analysis of Function









Working method for maximum and minimum:



If function $f(x)$ be given, find $f'(x)$ and equate it to zero. Solve this equation for x .
Let its roots be a_1, a_2, a_3 .

Find $f''(x)$ and hence find $f''(a_1) f''(a_2) f''(a_3)$.

If $f''(a_1) < 0$ we have a maximum at $x=a_1$ and maximum value= $f(a_1)$.

If $f''(a_1) > 0$ we have a minimum at $x=a_1$ and minimum value= $f(a_1)$.

If $f''(a_1) = 0$ Find $f'''(x)$ and hence $f'''(a_1)$.

If $f'''(a_1) \neq 0$ then there is neither maxima nor minima at $x=a_1$.

If $f'''(a_1) = 0$ Find $f^{iv}(x)$ and hence $f^{iv}(a_1)$.

Conclude as like as $f''(a_1)$ for the case $f^{iv}(a_1)$ is positive or negative or zero.

Working method (Contd):



Summary: $f'(a_1) = f''(a_1) = \dots = f^{n-1}(a_1) = 0$ and $f^n(a_1) \neq 0$

If n is odd then there will be neither maxima nor minima at $x=a_1$.

If n is even and if $f^n(a_1) > 0$ then we have a max. at $x=a_1$ and maximum value $= f(a_1)$.

If n is even and if $f^n(a_1) < 0$ then we have a min. at $x=a_1$ and minimum value $= f(a_1)$.

Find the maximum and minimum values of $y=x^5 -5x^4 +5x^3 -10$.



Given $y=x^5 -5x^4 +5x^3 -10$.

$$y' = 5x^4 - 20x^3 + 15x^2.$$

$$y'' = 20x^3 - 60x^2 + 30x.$$

$$y''' = 60x^2 - 120x + 30.$$

Setting $y' = 0$, we get

$$x^4 - 4x^3 + 3x^2 = 0$$

$$\Rightarrow x^2 (x^2 - 4x + 3) = 0$$

$$\Rightarrow x^2 (x - 3) (x - 1) = 0$$

$$\Rightarrow x = 0, 1, 3 \text{ [Critical point]}$$

$$\text{Now, } y''|_{x=1} = 20 \times 1^3 - 60 \times 1^2 + 30 \times 1 = -10 < 0.$$

Therefore, the given function is maximum when $x=1$

$$\text{and maximum value} = 1^5 - 5 \times 1^4 + 5 \times 1^3 - 10 = -9$$

Find the maximum and minimum values of $y=x^5 -5x^4 +5x^3 -10$.



Given $y=x^5 -5x^4 +5x^3 -10$.

$$y' = 5x^4 - 20x^3 + 15x^2.$$

$$y'' = 20x^3 - 60x^2 + 30x.$$

$$y''' = 60x^2 - 120x + 30.$$

$$\text{Now, } y''|_{x=3} = 20 \times 3^3 - 60 \times 3^2 + 30 \times 3 = 90 > 0.$$

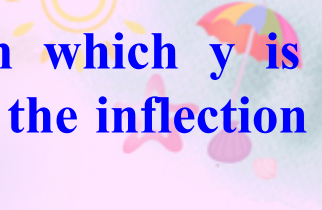
Therefore, the given function is minimum when $x=3$

$$\text{and minimum value} = 3^5 - 5 \times 3^4 + 5 \times 3^3 - 10 = -37$$

$$\text{Now, } y''|_{x=0} = 20 \times 0^3 - 60 \times 0^2 + 30 \times 0 = 0.$$

$$y'''|_{x=0} = 60 \times 0^2 - 120 \times 0 + 30 = 30 \neq 0.$$

Therefore, the given function is neither maximum nor minimum when $x=0$.



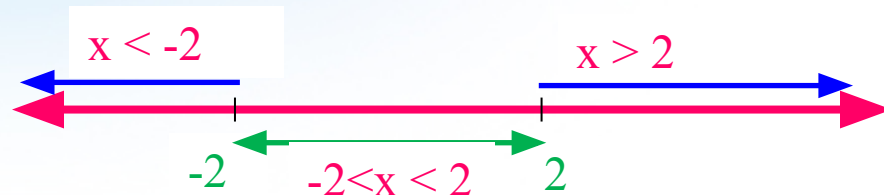
Find the critical points of $y=5+12x-x^3$. Identify the intervals on which y is increasing and decreasing, concave up and concave down. Also find the inflection points, if any, of the above function.

Given $y=5+12x-x^3$.

$$y' = 12 - 3x^2 = 3(4 - x^2) = 3(2 - x)(2 + x)$$

$$y'' = -6x$$

The sign analysis of y' is given below:

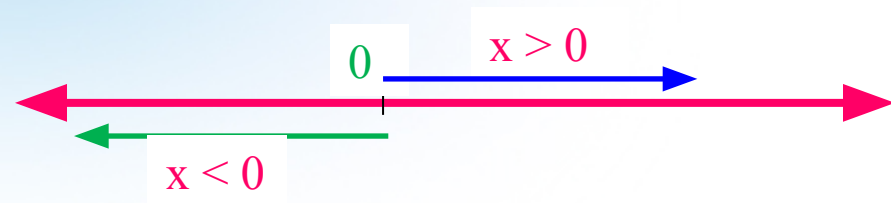


<i>Interval</i>	$(2+x)$	$(2-x)$	$f'(x)$	<i>Conclusion</i>
$x < -2$	-ve	+ve	-ve	$f(x)$ is decreasing on $(-\infty, -2]$
$-2 < x < 2$	+ve	+ve	+ve	$f(x)$ is increasing on $[-2, 2]$
$x > 2$	+ve	-ve	-ve	$f(x)$ is decreasing on $[2, \infty)$

$$y' = 12 - 3x^2$$

$$y'' = -6x$$

The sign analysis of y'' is given below:



<i>Interval</i>	$f''(x)$	<i>Conclusion</i>
$x < 0$	+ve	$f(x)$ is Concave up on $(-\infty, 0)$
$x > 0$	-ve	$f(x)$ is Concave down on $(0, \infty)$

From the above sign analysis of $f''(x)$ we see that $x=0$ is an inflection point of the given function.

Home Work



Find the maximum and minimum values of the following

- $y = 3x^5 - 25x^2 + 60x.$

- $y = x^6 - 12x^5 + 36x^4 + 4.$

- $y = (x^3 / 3) + ax^2 - 3a^2 x$

- $y = x^{1/x}.$

Find the critical points of the following function . Identify the intervals on which y is increasing and decreasing, concave up and concave down. Also find the inflection points, if any, of the above function.

- $y = 3x^4 + 4x^3 - 12x^2 + 2.$

- $y = x^6 - 12x^5 + 36x^4 + 4.$

- $y = x^5 - 5x^4 + 5x^3 - 10$

- $y = 3x^4 + 4x^3 - 12x^2 + 10.$



THANK YOU